Fundamental Trigonometric Identities

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Using the Pythagorean identities**

Given that $\sin\theta = \frac{3}{5}$ and θ lies in Quadrant II, find:

a $\cos\theta = -\frac{4}{5}$ **b** $\tan\theta = -\frac{3}{4}$ **c** $\sec\theta = -\frac{5}{4}$ **d** $\csc\theta = \frac{5}{3}$

(a) $= -\frac{4}{5}$; (b) $= -\frac{3}{4}$; (c) $= -\frac{5}{4}$; (d) $= \frac{5}{3}$

2 **[Foundational]** • **Target: Using the Pythagorean identities**

Given that $\tan\theta = 2$ and θ lies in Quadrant III, find $\sin\theta$ and $\cos\theta$ in exact simplified form.

$\sec^2\theta = 1 + \tan^2\theta = 5$, and in QIII $\cos\theta < 0$, so $\cos\theta = -\frac{\sqrt{5}}{5}$ and $\sin\theta = \tan\theta\cos\theta = -\frac{2\sqrt{5}}{5}$.

3 **[Foundational]** • **Target: Simplifying and verifying**

Simplify each expression to a single trigonometric function:

a $\frac{1-\cos^2 x}{\sin x} = \frac{\sin^2 x}{\sin x} = \sin x$

b $\sec x\cot x = \frac{1}{\cos x} \cdot \frac{\cos x}{\sin x} = \csc x$

c $\tan x\csc x = \frac{\sin x}{\cos x} \cdot \frac{1}{\sin x} = \sec x$

(a) $= \frac{\sin^2 x}{\sin x} = \sin x$; (b) $= \frac{1}{\cos x} \cdot \frac{\cos x}{\sin x} = \csc x$; (c) $= \frac{\sin x}{\cos x} \cdot \frac{1}{\sin x} = \sec x$

4 **[Foundational]** • **Target: Simplifying and verifying**

Verify the identity $(1 + \tan^2 x)\cos^2 x = 1$.

$(1 + \tan^2 x)\cos^2 x = \sec^2 x\cos^2 x = \frac{1}{\cos^2 x}\cos^2 x = 1.$

5 **[Foundational]** • **Target: Simplifying and verifying**

Verify the identity $\frac{\sin x}{1+\cos x} = \frac{1-\cos x}{\sin x}$.

Cross-multiplying view: $\sin^2 x = (1 + \cos x)(1 - \cos x) = 1 - \cos^2 x$, which is the Pythagorean identity.

Equivalently, multiply the left side by $\frac{1-\cos x}{1-\cos x}$ to obtain $\frac{\sin x(1-\cos x)}{1-\cos^2 x} = \frac{\sin x(1-\cos x)}{\sin^2 x} = \frac{1-\cos x}{\sin x}$.

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6 [Foundational] • Target: Quotient and reciprocal identities

Rewrite each expression using only $\sin x$ and $\cos x$, then simplify:

a $\cot x \sin x$ $= \frac{\cos x}{\sin x} \cdot \sin x = \cos x$

b $\frac{\sec x}{\csc x}$ $= \frac{1/\cos x}{1/\sin x} = \frac{\sin x}{\cos x} = \tan x$

c $1 - (\sin^2 x \csc^2 x)\sin^2 x$ $1 - \sin^2 x = \cos^2 x$ (since $\csc^2 x \sin^2 x = 1$)

(a) $= \frac{\cos x}{\sin x} \cdot \sin x = \cos x$; (b) $= \frac{1/\cos x}{1/\sin x} = \frac{\sin x}{\cos x} = \tan x$; (c) $1 - \sin^2 x = \cos^2 x$ (since $\csc^2 x \sin^2 x = 1$)

7 [Foundational] • Target: Quotient and reciprocal identities

Simplify $\frac{\cos x}{1 - \sin x} - \tan x$ to a single term.

Combine over $\cos x(1 - \sin x)$: $\frac{\cos^2 x - \sin x(1 - \sin x)}{\cos x(1 - \sin x)} = \frac{\cos^2 x + \sin^2 x - \sin x}{\cos x(1 - \sin x)} = \frac{1 - \sin x}{\cos x(1 - \sin x)} = \sec x$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: More identity verifications

Verify the identity $\sec^2 x - \tan^2 x = 1$ using the Pythagorean identity for tangent and secant.

The identity $1 + \tan^2 x = \sec^2 x$ (dividing $\sin^2 x + \cos^2 x = 1$ by $\cos^2 x$) rearranges directly to $\sec^2 x - \tan^2 x = 1$.

9 [Intermediate] • Target: More identity verifications

Verify the identity $\frac{1}{1 - \cos x} + \frac{1}{1 + \cos x} = 2\csc^2 x$.

Combine over a common denominator: $\frac{(1 + \cos x) + (1 - \cos x)}{(1 - \cos x)(1 + \cos x)} = \frac{2}{1 - \cos^2 x} = \frac{2}{\sin^2 x} = 2\csc^2 x$.

10 [Intermediate] • Target: More identity verifications

Show that $(\sin x + \cos x)^2 = 1 + 2\sin x \cos x$.

Expand: $(\sin x + \cos x)^2 = \sin^2 x + 2\sin x \cos x + \cos^2 x = (\sin^2 x + \cos^2 x) + 2\sin x \cos x = 1 + 2\sin x \cos x$.

11 [Intermediate] • Target: More identity verifications

Given that $\cos\theta = -\frac{8}{17}$ and θ lies in Quadrant III, find $\sin\theta$, $\tan\theta$, and $\cot\theta$ in exact simplified form.

$\sin^2\theta = 1 - \frac{64}{289} = \frac{225}{289}$, and in QIII $\sin\theta < 0$: $\sin\theta = -\frac{15}{17}$. $\tan\theta = \frac{-15/17}{-8/17} = \frac{15}{8}$. $\cot\theta = \frac{8}{15}$.

12 [Intermediate] • Target: Identities with sums and products

Verify the identity $\sin x \cos x(\tan x + \cot x) = 1$.

$\sin x \cos x\left(\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x}\right) = \sin^2 x + \cos^2 x = 1$.

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13 [Intermediate] • Target: Identities with sums and products

Simplify $(\sec x - \tan x)(\sec x + \tan x)$ to a constant.

This is a difference of squares: $\sec^2 x - \tan^2 x = 1$ (a Pythagorean identity).

14 [Intermediate] • Target: Identities involving all three Pythagorean forms

State all three forms of the Pythagorean identity (for sine/cosine, tangent/secant, and cotangent/cosecant).

$\sin^2 x + \cos^2 x = 1$; $1 + \tan^2 x = \sec^2 x$; $1 + \cot^2 x = \csc^2 x$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Identities involving all three Pythagorean forms • Cross-Lesson: 8.01 Rational Functions and Asymptotes

Given $\cot\theta = \frac{5}{12}$ and θ in Quadrant I, find $\csc\theta$ and $\sin\theta$ using the cotangent-cosecant identity.

$\csc^2\theta = 1 + \frac{25}{144} = \frac{169}{144}$, and in QI $\csc\theta > 0$: $\csc\theta = \frac{13}{12}$. $\sin\theta = \frac{1}{\csc\theta} = \frac{12}{13}$.

16 [Challenging] • Target: Simplifying more complex expressions • Cross-Lesson: 8.02 Graphing Rational Functions

Simplify $\frac{\sin^3 x + \cos^3 x}{\sin x + \cos x}$ using the sum-of-cubes factoring pattern.

Factor as a sum of cubes: $\sin^3 x + \cos^3 x = (\sin x + \cos x)(\sin^2 x - \sin x\cos x + \cos^2 x)$. Dividing: $\frac{(\sin x + \cos x)(1 - \sin x\cos x)}{\sin x + \cos x} = 1 - \sin x\cos x$.

17 [Challenging] • Target: Verifying identities with a common denominator strategy • Cross-Lesson: 8.03 Introduction to Limits

Verify $\tan x + \cot x = \sec x\csc x$.

$\tan x + \cot x = \frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} = \frac{\sin^2 x + \cos^2 x}{\sin x\cos x} = \frac{1}{\sin x\cos x} = \sec x\csc x$.

18 [Challenging] • Target: Verifying identities with a common denominator strategy • Cross-Lesson: 9.01 Arithmetic Sequences and Series

Verify $\frac{\sin x}{\csc x} + \frac{\cos x}{\sec x} = 1$.

$\frac{\sin x}{\csc x} = \sin x \cdot \sin x = \sin^2 x$; $\frac{\cos x}{\sec x} = \cos x \cdot \cos x = \cos^2 x$. Sum: $\sin^2 x + \cos^2 x = 1$.

19 [Challenging] • Target: Numeric identity verification (checking a specific angle) • Cross-Lesson: 9.02 Geometric Sequences and Series

Verify the identity $1 + \tan^2 x = \sec^2 x$ numerically at $x = \frac{\pi}{6}$ by computing both sides.

$\tan\frac{\pi}{6} = \frac{1}{\sqrt{3}}$: LHS $= 1 + \frac{1}{3} = \frac{4}{3}$. $\sec\frac{\pi}{6} = \frac{1}{\cos(\pi/6)} = \frac{2}{\sqrt{3}}$: RHS $= \frac{4}{3}$. Both sides match, confirming the identity at this angle (though this alone doesn't prove it for all x).

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20 [Challenging] • Target: Identities with squared expressions • Cross-Lesson: 9.03 The Binomial Theorem

Simplify $(1 - \sin^2 x)(1 + \tan^2 x)$ to a constant.

$(1 - \sin^2 x) = \cos^2 x$ and $(1 + \tan^2 x) = \sec^2 x = \frac{1}{\cos^2 x}$. Product: $\cos^2 x \cdot \frac{1}{\cos^2 x} = 1$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Reciprocal and Quotient Identities

Simplify the expression $\tan x \cdot \csc x$.

a. $\sec x$ ✓ **b.** $\cos x$

c. $\sin x$ **d.** $\cot x$

$\tan x \cdot \csc x = \frac{\sin x}{\cos x} \cdot \frac{1}{\sin x} = \frac{1}{\cos x} = \sec x$.

22 [Foundational] • Target: Reciprocal and Quotient Identities

Simplify $\frac{\cos x}{\cot x}$.

a. $\sin x$ ✓ **b.** $\cos^2 x$

c. $\tan x$ **d.** $\csc x$

$\frac{\cos x}{\cos x/\sin x} = \cos x \cdot \frac{\sin x}{\cos x} = \sin x$.

23 [Foundational] • Target: Reciprocal and Quotient Identities

If $\sin\theta = -3/5$ and $\tan\theta > 0$, find $\cos\theta$.

a. $-4/5$ ✓ **b.** $4/5$

c. $-3/4$ **d.** $5/4$

$\sin < 0$ and $\tan > 0 \Rightarrow$ Quadrant III. In QIII, $\cos\theta = -\sqrt{1 - (-3/5)^2} = -\sqrt{16/25} = -4/5$.

24 [Foundational] • Target: Pythagorean Identities

Which identity is directly derived by dividing $\sin^2\theta + \cos^2\theta = 1$ by $\cos^2\theta$?

a. $1 + \tan^2\theta = \sec^2\theta$ ✓ **b.** $1 + \cot^2\theta = \csc^2\theta$

c. $\sin^2\theta = 1 - \cos^2\theta$ **d.** $\tan^2\theta + \cot^2\theta = 1$

$\frac{\sin^2\theta}{\cos^2\theta} + \frac{\cos^2\theta}{\cos^2\theta} = \frac{1}{\cos^2\theta} \Rightarrow \tan^2\theta + 1 = \sec^2\theta$.

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25 [Foundational] • Target: Pythagorean Identities

Simplify $(\sec\theta - 1)(\sec\theta + 1)$.

a. $\tan^2\theta$ ✓

b. $\cot^2\theta$

c. $\cos^2\theta$

d. 1

Difference of squares: $\sec^2\theta - 1 = \tan^2\theta$.

26 [Foundational] • Target: Pythagorean Identities

Simplify $\frac{1-\cos^2 x}{\sin x}$.

a. $\sin x$ ✓

b. $\cos x$

c. $\csc x$

d. 1

$\frac{\sin^2 x}{\sin x} = \sin x$.

27 [Foundational] • Target: Even/Odd Identities and Symmetry

Which trigonometric functions are even functions ($f(-\theta) = f(\theta)$)?

a. $\cos\theta$ **and** $\sec\theta$ ✓

b. $\sin\theta$ and $\csc\theta$

c. $\tan\theta$ and $\cot\theta$

d. All six functions

Only cosine and its reciprocal secant are even: $\cos(-\theta) = \cos\theta$ and $\sec(-\theta) = \sec\theta$.

28 [Foundational] • Target: Even/Odd Identities and Symmetry

Simplify $\sin(-\theta) \cdot \cot(-\theta)$.

a. $\cos\theta$ ✓

b. $-\cos\theta$

c. $\sin\theta$

d. $-\sin\theta$

$\sin(-\theta) = -\sin\theta$ and $\cot(-\theta) = -\cot\theta$. Product $= (-\sin\theta)(-\cot\theta) = \sin\theta \cdot \frac{\cos\theta}{\sin\theta} = \cos\theta$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Even/Odd Identities and Symmetry

Simplify $\frac{\tan(-x)}{\sin(-x)}$.

a. $\sec x$ ✓

b. $-\sec x$

c. $\cos x$

d. $-\cos x$

$\frac{-\tan x}{-\sin x} = \frac{\tan x}{\sin x} = \frac{\sin x/\cos x}{\sin x} = \frac{1}{\cos x} = \sec x$.

30 [Intermediate] • Target: Verifying Identities Algebraically

Prove $\frac{\sin x}{1+\cos x} = \frac{1-\cos x}{\sin x}$. What technique verifies this algebraically?

a. **Multiply numerator and denominator of the left side by the conjugate** $(1-\cos x)$ ✓

b. Square both sides

c. Take the derivative

d. Cross multiply and set to 0

$\frac{\sin x(1-\cos x)}{(1+\cos x)(1-\cos x)} = \frac{\sin x(1-\cos x)}{1-\cos^2 x} = \frac{\sin x(1-\cos x)}{\sin^2 x} = \frac{1-\cos x}{\sin x}$.

31 [Intermediate] • Target: Verifying Identities Algebraically

Simplify $\frac{\sec x - \cos x}{\sin x}$.

a. $\tan x$ ✓

b. $\cot x$

c. $\sec x$

d. $\sin x$

$\frac{1/\cos x - \cos x}{\sin x} = \frac{(1-\cos^2 x)/\cos x}{\sin x} = \frac{\sin^2 x}{\cos x \sin x} = \frac{\sin x}{\cos x} = \tan x$.

32 [Intermediate] • Target: Verifying Identities Algebraically

Simplify $(1+\tan^2\theta)(1-\sin^2\theta)$.

a. 1 ✓

b. $\sec^2\theta$

c. $\tan^2\theta$

d. $\cos^2\theta$

$(1+\tan^2\theta) = \sec^2\theta$ and $(1-\sin^2\theta) = \cos^2\theta$. Product $= \sec^2\theta \cdot \cos^2\theta = 1$.

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33 [Intermediate] • Target: Even/Odd Identities and Symmetry

If $\cos\theta = \frac{77}{85}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.

a. $\frac{77}{85}$

b. $\frac{36}{85}$ ✓

c. $-\frac{36}{85}$

d. $\frac{89}{170}$

$\sin\theta = \sqrt{1-\cos^2\theta} = \sqrt{1-\frac{5929}{7225}} = \frac{36}{85}$.

34 [Intermediate] • Target: Verifying Identities Algebraically

If $\cos\theta = \frac{80}{89}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.

a. $\frac{80}{89}$

b. $-\frac{39}{89}$

c. $\frac{39}{89}$ ✓

d. $\frac{479}{890}$

$\sin\theta = \sqrt{1-\cos^2\theta} = \sqrt{1-\frac{6400}{7921}} = \frac{39}{89}$.

35 [Intermediate] • Target: Using the Pythagorean identities

If $\cos\theta = \frac{40}{41}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.

a. $\frac{131}{410}$

b. $\frac{40}{41}$

c. $\frac{9}{41}$ ✓

d. $-\frac{9}{41}$

$\sin\theta = \sqrt{1-\cos^2\theta} = \sqrt{1-\frac{1600}{1681}} = \frac{9}{41}$.

36 [Intermediate] • Target: Simplifying and verifying

If $\cos\theta = \frac{21}{29}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.

a. $-\frac{20}{29}$

b. $\frac{229}{290}$

c. $\frac{20}{29}$ ✓

d. $\frac{21}{29}$

$\sin\theta = \sqrt{1-\cos^2\theta} = \sqrt{1-\frac{441}{841}} = \frac{20}{29}$.

37 [Intermediate] • Target: Quotient and reciprocal identities

If $\cos\theta = \frac{12}{13}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.

a. $\frac{63}{130}$

b. $\frac{12}{13}$

c. $-\frac{5}{13}$

d. $\frac{5}{13}$ ✓

$\sin\theta = \sqrt{1-\cos^2\theta} = \sqrt{1-\frac{144}{169}} = \frac{5}{13}$.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: More identity verifications • Cross-Lesson: 9.01 Arithmetic Sequences and Series

If $\cos\theta = \frac{15}{17}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.

a. $\frac{15}{17}$

b. $\frac{97}{170}$

c. $\frac{8}{17}$ ✓

d. $-\frac{8}{17}$

$\sin\theta = \sqrt{1-\cos^2\theta} = \sqrt{1-\frac{225}{289}} = \frac{8}{17}$.

39 [Challenging] • Target: Identities with sums and products • Cross-Lesson: 9.02 Geometric Sequences and Series

If $\cos\theta = \frac{24}{25}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.

a. $\frac{24}{25}$

b. $\frac{19}{50}$

c. $\frac{7}{25}$ ✓

d. $-\frac{7}{25}$

$\sin\theta = \sqrt{1-\cos^2\theta} = \sqrt{1-\frac{576}{625}} = \frac{7}{25}$.

40 [Challenging] • Target: Identities involving all three Pythagorean forms • Cross-Lesson: 9.03 The Binomial Theorem

If $\cos\theta = \frac{4}{5}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.

a. $\frac{4}{5}$

b. $-\frac{3}{5}$

c. $\frac{3}{5}$ ✓

d. $\frac{7}{10}$

$\sin\theta = \sqrt{1-\cos^2\theta} = \sqrt{1-\frac{16}{25}} = \frac{3}{5}$.

41 [Challenging] • Target: Simplifying more complex expressions • Cross-Lesson: 1.01 Functions and Function Notation

If $\sin\theta = \frac{33}{65}$, find $\cos^2\theta$ using the Pythagorean identity.

a. $\frac{1089}{4225}$

b. $\frac{56}{65}$

c. $\frac{3136}{4225}$ ✓

d. $\frac{7117}{8450}$

$\cos^2\theta = 1-\sin^2\theta = 1-\frac{1089}{4225} = \frac{3136}{4225}$.

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42 [Challenging] • Target: Verifying identities with a common denominator strategy • Cross-Lesson: 1.02 Domain and Range

If $\sin\theta = \frac{48}{73}$, find $\cos^2\theta$ using the Pythagorean identity.

a. $\frac{3025}{5329}$ ✓

b. $\frac{2304}{5329}$

c. $\frac{55}{73}$

d. $\frac{35579}{53290}$

$\cos^2\theta = 1 - \sin^2\theta = 1 - \frac{2304}{5329} = \frac{3025}{5329}$.

43 [Challenging] • Target: Numeric identity verification (checking a specific angle) • Cross-Lesson: 1.03 Average Rate of Change

If $\sin\theta = \frac{36}{85}$, find $\cos^2\theta$ using the Pythagorean identity.

a. $\frac{5929}{7225}$ ✓

b. $\frac{1296}{7225}$

c. $\frac{77}{85}$

d. $\frac{13303}{14450}$

$\cos^2\theta = 1 - \sin^2\theta = 1 - \frac{1296}{7225} = \frac{5929}{7225}$.

44 [Challenging] • Target: Identities with squared expressions • Cross-Lesson: 2.01 Right-Triangle Trigonometry

If $\sin\theta = \frac{39}{89}$, find $\cos^2\theta$ using the Pythagorean identity.

a. $\frac{1521}{7921}$

b. $\frac{80}{89}$

c. $\frac{6400}{7921}$ ✓

d. $\frac{71921}{79210}$

$\cos^2\theta = 1 - \sin^2\theta = 1 - \frac{1521}{7921} = \frac{6400}{7921}$.

45 [Challenging] • Target: Reciprocal and Quotient Identities • Cross-Lesson: 2.02 Radian Measure and Arc Length

If $\sin\theta = \frac{11}{61}$, find $\cos^2\theta$ using the Pythagorean identity.

a. $\frac{121}{3721}$

b. $\frac{39721}{37210}$

c. $\frac{3600}{3721}$ ✓

d. $\frac{60}{61}$

$\cos^2\theta = 1 - \sin^2\theta = 1 - \frac{121}{3721} = \frac{3600}{3721}$.

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